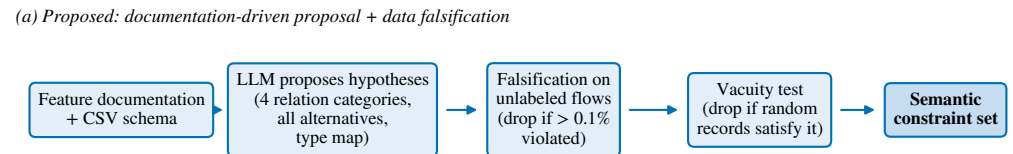
(a) Proposed: documentation-driven proposal + data falsification
Feature documentation + CSV schema
LLM proposes hypotheses (4 relation categories, all alternatives, type map)
Falsification on unlabeled flows (drop if > 0.1% violated)
Vacuity test (drop if random records satisfy it)
Semantic constraint set
(b) Baselines: constraints learned from data
Observed flows (benign)
Statistical learning (value ranges, regression pairs) or template mining (Daikon-style)
Data-learned constraint set